Kimball Wang write up
Math 104 HW5

Problem 2

a) Suppose f is bounded and real, and suppose f^2 is Riemann integrable (with respect to $\alpha(x)=x$). Does it follow that f is Riemann-integrable?

Solution

No

Define $f(x)=\begin{cases} -1, & x\notin\mathbb{Q},\ x\in[a,b] \\ 1, & x\in\mathbb{Q},\ x\in[a,b]\end{cases}$

$|f(x)|\leq 2 \quad \forall x\in[a,b]$

f is bounded

$f(x)\in\{-1,1\}\ \forall x\in[a,b]$

f is real

f^2, being the constant function 1, is integrable.

Consider any partition P of $[a,b]$

$$P=\{a=t_0<t_1<t_2<\cdots<t_{n-1}<t_n=b\}$$

$\mathbb{Q}$ is dense in $\mathbb{R}$, and
$\mathbb{R}-\mathbb{Q}$ is dense in $\mathbb{R}$

$M_i=1 \quad \forall i\in\{1,2,\ldots,n\}$
$m_i=-1 \quad \forall i\in\{1,2,\ldots,n\}$

Hence

$$U(f,P)=\sum_{i=1}^{n}M_i\Delta t_i=\sum_{i=1}^{n}\Delta t_i$$
$$=b-a$$

$$L(f,P)=\sum_{i=1}^{n}m_i\Delta t_i=-\sum_{i=1}^{n}\Delta t_i$$
$$=-(b-a)$$
$$=a-b$$

$U(f)=\inf U(f,P)=b-a$
$L(f)=\sup L(f,P)=a-b$

$U(f) \neq L(f) \Rightarrow f$ is not Riemann Integrable

b) Suppose f is bounded and real, and suppose f^3 is Riemann integrable (with respect to $\alpha(x) = x$). Does it follow that f is Riemann-integrable?

Solution

First we bring some definition and theorem from Rudin Book

Definition 6.1 Let $[a,b]$ be a given interval. By a partition P of $[a,b]$ we mean a finite set of points $x_0, x_1, \dots, x_n$, where

$$a = x_0 \leq x_1 \leq \cdots \leq x_{n-1} \leq x_n = b$$

we write

$$\Delta x_i = x_i - x_{i-1} \quad (i = 1, \dots, n).$$

Now suppose f is a bounded real function defined on $[a,b]$. Corresponding to each partition P of $[a,b]$ we put

$$M_i = \sup f(x) \quad (x_{i-1} \leq x \leq x_i),$$

$$m_i = \inf f(x) \quad (x_{i-1} \leq x \leq x_i),$$

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i,$$

$$L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i,$$

2

Theorem 6.6 $f \in \mathcal{R}(\alpha)$ on $[a,b]$ if and only if for every $\varepsilon > 0$ there exists a partition P such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

Theorem 6.11 Suppose $f \in \mathcal{R}(\alpha)$ on $[a,b]$, $m \leq f \leq M$, ϕ is continuous on $[m, M]$, and $h(x) = \phi(f(x))$ on $[a,b]$. Then $h \in \mathcal{R}(\alpha)$ on $[a,b]$.

Proof Choose $\varepsilon > 0$. Since ϕ is uniformly continuous on $[m, M]$, there exists $\delta > 0$ such that $\delta < \varepsilon$ and $|\phi(s) - \phi(t)| < \varepsilon$ if $|s - t| \leq \delta$ and $s, t \in [m, M]$.

Since $f \in \mathcal{R}(\alpha)$, there is a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \delta^2$$

Let M_i, m_i have the same meaning as in Definition 6.1, and let M_i^*, m_i^* be the analogous numbers for h. Divide the numbers $1, \ldots, n$ into two classes:

$i \in A$ if $M_i - m_i < \delta$, $i \in B$ if $M_i - m_i \geq \delta$

For $i \in A$, our choice of δ shows that

$$M_i^* - m_i^* \leq \varepsilon$$

For $i \in B$, $M_i^* - m_i^* \leq 2K$, where $K = \sup |\phi(t)|$, $m \leq t \leq M$. By

$$U(P, f, \alpha) - L(P, f, \alpha) < \delta^2$$

Let we have

$$\delta \sum_{i \in B} \Delta\alpha_i \leq \sum_{i \in B} (M_i - m_i) \Delta\alpha_i < \delta^2$$

3 so that $\sum_{i \in B} \Delta\alpha_i < \delta$. It follows that

$$\begin{aligned} U(P, h, \alpha) - L(P, h, \alpha) &= \sum_{i \in A} (M_i^* - m_i^*) \Delta\alpha_i + \sum_{i \in B} (M_i^* - m_i^*) \Delta\alpha_i \\ &\leq \varepsilon[\alpha(b) - \alpha(a)] + 2K\delta \\ &< \varepsilon[\alpha(b) - \alpha(a) + 2K] \end{aligned}$$

Since ε was arbitrary, Theorem 6.6 implies that $h \in \mathcal{R}(\alpha)$.

Here is the actual homework solution

Proof Suppose $f^3 \in \mathcal{R}$ on $[a, b]$

Since f is bounded

we have that $m \le f \le M$
for some m and M
The function $\phi(x) = x^{1/3}$ is continuous on all of $\mathbb{R}$
$\Rightarrow$ The function $\phi(x) = x^{1/3}$ is continuous on $[m, M]$
Hence, $h(x) = \phi((f(x))^3) = f(x) \in \mathcal{R}$ on $[a, b]$
$\because$ Theorem 6.11

c) Suppose that f is infinitely differentiable everywhere. Suppose that there is some $L > 0$ such that $|f^{(n)}(x)| < L$ for all n and $x \in \mathbb{R}$. Further suppose that $f(\frac{1}{n}) = 0$ for all $n \in \mathbb{N}$. Show that $f(x) = 0$ everywhere

Proof

First we show $f^{(n)}(0) = 0 \quad \forall n \in \mathbb{N}$

for all $n \in \mathbb{N}$

consider the interval $I_n = [\frac{1}{n}, \frac{1}{n+1}]$

$f(\frac{1}{n}) = 0$, $f(\frac{1}{n+1}) = 0$

By mean value theorem,

$$\frac{f(\frac{1}{n+1}) - f(\frac{1}{n})}{\frac{1}{n+1} - \frac{1}{n}} = f'(c_n)$$

for some $c_n \in (\frac{1}{n}, \frac{1}{n+1})$

$$\Rightarrow \frac{0 - 0}{\frac{1}{n+1} - \frac{1}{n}} = f'(c_n)$$

4 $\Rightarrow 0 = f'(c_n)$

As $n \to \infty$ $\lim_{n\to\infty} \frac{1}{n} = 0 = \lim_{n\to\infty} \frac{1}{n+1}$

$\frac{1}{n} < c_n < \frac{1}{n+1}$

By squeeze theorem

As $n \to \infty$ $c_n \to 0$

Hence $f'(0) = 0$

apply the same argument to $I_n' = [c_n, c_{n+1}]$

$f'(c_n) = 0$, $f'(c_{n+1}) = 0$

By Mean Value Theorem

$f''(c_n') = 0$ for some $c_n' \in (c_n, c_{n+1})$

As $n \to \infty$ $c_n \to 0$ $c_{n+1} \to 0$

$c_n < c_n' < c_{n+1}$

By squeeze theorem As $n \to \infty$ $c_n' \to 0$

Hence $f^{(2)}(0)=0$

For inductive step

Assume $c_n^m \to 0$ as $n\to\infty$ $\quad c_n^m \in (\frac{1}{n}, \frac{1}{n+1})$

Assume $c_{n+1}^m \to 0$ as $n\to\infty$ $\quad c_{n+1}^m \in (\frac{1}{n+1}, \frac{1}{n+2})$

for some $m \in \mathbb{N}$

apply the same argument to

interval $I_n^{m+1} = [c_n^m, c_{n+1}^m]$

$f^{(m+1)}(c_n^m)=0$, $f^{(m+1)}(c_{n+1}^m)=0$

By mean value theorem

$f^{(m+2)}(c_n^{m+1})=0$ for some $c_n^{m+1} \in (c_n^m, c_{n+1}^m)$

As $n\to\infty$ $\quad c_n^m \to 0$, $c_{n+1}^m \to 0$

$c_n^m < c_n^{m+1} < c_{n+1}^m$

$\Rightarrow$ By squeeze theorem

As $n\to\infty$ $c_n^{m+1} \to 0$

$\Rightarrow f^{(m+2)}(0)=0$

Hence by Principle of Mathematical Induction

$f^{(m)}(0)=0 \quad \forall m \in \mathbb{N}$

Now we use Taylor Theorem

to show $f(s)=0 \quad \forall s \in \mathbb{R}$

By Taylor Theorem,

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta-\alpha)^n$$

5 $$P(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(\beta-\alpha)^k \quad , \alpha, \beta \in \mathbb{R}, \; x \in (\alpha, \beta)$$

If we let $\alpha = 0$

$$P(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(0)}{k!}(\beta)^k = 0 \quad \because f^{(k)}(0)=0 \text{ proved above}$$

$$f(\beta) = \frac{f^{(n)}(x)}{n!}(\beta)^n$$

$$|f(\beta)| = \left|\frac{f^{(n)}(x)}{n!}(\beta)^n\right|$$

$$\leq \frac{L}{n!}(\beta)^n \quad \forall n$$

$$\lim_{n\to\infty} \frac{\beta^n}{n!} = \lim_{n\to\infty} \left(\frac{\beta}{n}\right)\cdot\left(\frac{\beta}{n-1}\right)\cdots\left(\frac{\beta}{1}\right) = 0$$

Hence $\lim_{n\to\infty} \frac{L}{n!}(\beta)^n = 0$

$$\Rightarrow \lim_{n\to\infty} |f(\beta)| \leq 0$$

$$\Rightarrow \lim_{n\to\infty} |f(\beta)| = 0$$

$$\Rightarrow \lim_{n\to\infty} f(\beta) = 0$$

$$\Rightarrow f(\beta) = 0 \quad \forall \beta \in \mathbb{R}$$

(d) Define the integral

$$\int_a^\infty f\,dx := \lim_{b\to\infty} \int_a^b f\,dx$$

If it exists, we say it converges.

Suppose that $f(x) \geq 0$ and f decreases monotonically on $[1,\infty)$. Show that $\int_1^\infty f\,dx$ converges if and only if $\sum_{n=1}^\infty f(n)$ converges.

Proof

Since f is nonnegative and decreases monotonically on $[1,\infty)$, we have

6 $$\sum_{n=2}^{K} f(n) \leq \int_1^K f(x)\,dx \leq \sum_{n=1}^{K-1} f(n)$$

Let $g(K) = \int_1^K f(x)\,dx$

If $g(K)$ converges, then $g(K)$ is a nondecreasing function which tends to a limit, and hence $g(K)$ is a bounded nondecreasing sequence.

Let $S_K = \sum_{n=2}^{K} f(n)$

$$\Rightarrow S_K \leq g(K) \quad \forall K \in \mathbb{N},\ K \geq 2$$

$g(K)$ converge $\Rightarrow S_K$ converge

Hence $\int_1^\infty f\,dx$ converge $\Rightarrow \sum_{n=1}^\infty f(n)$ converge

For the converse

If $g(k)$ diverges

then $g(k) \to +\infty$ as $k \to +\infty$

Since $g(k) \leq \sum_{n=1}^{k-1} f(n)$

$\Rightarrow$ if $g(k) \to +\infty$ as $k \to +\infty$

then $\sum_{n=1}^{k-1} f(n) \to +\infty$ as $k \to +\infty$

$\Rightarrow \sum_{n=1}^{\infty} f(n) \to +\infty$

Hence $g(k)$ diverges as $k \to +\infty$

$\Rightarrow \sum_{n=1}^{\infty} f(n)$ diverges

Hence $\sum_{n=1}^{\infty} f(n)$ converges

$\Rightarrow g(k)$ converges as $k \to +\infty$

Hence $\sum_{n=1}^{\infty} f(n)$ converges

$\Rightarrow \int_1^{\infty} f\,dx$ converge

7